

Dancing Machaut's *Chansons baladées*: Unfixed Forms and Mensural Missteps

Social dancing was an integral part of both courtly and country life in medieval France and beyond.¹ Indeed, “dance was ever-present in the lives of western Europeans from the late Medieval period to the middle of the eighteenth century” (Nevile 2008a, 2). The *carole*—after the Old French verb *caroler*, to dance—was the simplest and most common type of dance, well established in France by the mid-twelfth century (Mullally 1986, 224).² Medieval French writings on a wide range of topics mention the *carole* and illuminated romances provide numerous depictions of it, such as that of **Example 1** from the *Roman d’Alexandre*.³ From these kinds of evidence, modern scholars have reconstructed the basic movement of the fourteenth-century *carole*. What remains is to attempt matching those dance steps to music we know to have been danced and, in turn, to speculate about how the structure of dance music may have encoded dance conventions. Alas, no European dance manuals exist before the fifteenth

¹ For a rich and broad sweep of dance’s importance to western European society—artistic, political, and intellectual—from 1250-1750, see Nevile 2008a and 2008b. Chaganti 2018 details how the central position of dance in late medieval society informed the creation and reception of lyric poetry. Page 1989, esp. Ch. 5, views the French *carole*’s social context from the perspective of an evolving clergy. Arcangeli 1994 reviews the colorful denunciation of dance by medieval clergy and moralists, not limited to claims of adultery, gluttony, paganism, lechery, and disregard for the Sabbath.

² Also *karole* and *querole*. Latin: *chorea*. The term was used both then and now to refer to the dance itself and to the music that accompanied dancing in a *carole*. Specific musical genres danced in this way will be clarified below.

³ Mullally 2011 discusses the etymology, choreography, and social participants of this simple dance without detailed analysis of musical structure. Greene 1977 offers a wealth of *carole* lyrics without music, in addition to summaries of the genre and of medieval English dancing.

century (Nevile 2008a, 7).⁴ While securely matching fourteenth-century music to dance may remain elusive, this essay takes the initial step of identifying basic norms. I then consider some complications that *ars nova* metrical innovations can impose upon a simple dance pattern and further explore how the formal organization of *formes fixes* lyrics can dislocate a dance's regularity. In doing so, I extend dance step notation as a tool for the analysis of metrical grouping in Machaut's more complex musical settings. This abstracted methodology is applicable to fourteenth-century song more broadly, including cases where genre, complexity, or polyphonic accompaniment likely excluded a particular song from being danced in its time. In whole, this study aims to foster a heightened awareness among modern interpreters of the dance roots of a considerable body of medieval French song and example how those roots can continue to shape our reading and performance of that repertoire today.

⁴ See Ward 1976 and Dean Smith 1937 for accounts of a fifteenth-century handbook on the basse danse that presents step patterns for dancing and music. Little 1975 discusses a Baroque dance manual more closely matched to given music, including implications for analysis and musical performance.



Example 1. dancing the *carole*; detail from MS Bodley 264, f. 172v (1338-44)

Unfortunately for scholars today, the *carole* was so simple that its proponents left very scant record of how to do it. The most revealing extant sources are condemnations of *carole* dancing by clergy and other moralists. We read in *Le Mireour du monde* (anonymous fourteenth century), for example, that “It is obvious that *caroles* are processions to the Devil, because they turn to the left. Of which Holy Scripture says: ‘God approves the ways that turn to the right; those that turn to the left are perverse and bad, and God hates them’” (Mullally

2011, 47).⁵ Or more descriptively in a sermon by Jacques de Vitry (before 1240) that the “*chorea* is a circle whose center is the Devil, and in it all turn to the left, because all are heading towards everlasting death. When foot is pressed to foot or the hand of a woman is touched by the hand of a man, there the fire of the Devil is kindled” (Mullally 2011, 49). The Latin and French verb *ferir*, common to many descriptions of caroling, suggests that after the left foot advanced, the right foot was brought together with a strike. From pairing such statements with a multitude of fourteenth-century illustrations, we know the *carole* to be a round dance proceeding clockwise with hands held. The typical participants were youthful women and men in alternation, when men were involved at all.⁶ The music of the *carole* was song sung by the dancers themselves, on rare occasion accompanied by a percussion instrument keeping the beat (Mullally 2011, 88).⁷ The *carole* was thus not a concert—music projected outwards for auditors—but song and dance for the entertainment (and often courtship) of the performers themselves. The heart of these songs was the refrain; simple, catchy melodies sung on repeat by the choir of dancers. The verses were sung by a soloist from among the group, and each

⁵ The same particularity with direction is also found in Dante’s *Inferno* (1320), in which the Pilgrim’s spiraling descent through Hell turns only to the left (Canto XIV, 125-7).

⁶ Nevile 2008a, 30, gives a recent summary of this widely-accepted description. That the *carole* could also be a line dance progressing through the streets of town is noted by Page 1990, 115, and McGee 1989, 507. The formation of the dance has no bearing on the approach here.

⁷ Nevile 2008b, 303, notes that Giotto’s early fourteenth-century fresco in Padua’s Scrovegni Chapel depicts a tambourine player accompanying the *carole*.

successive dance struck up by a new lead singer.⁸ An exemplar for this role can be found in Dame Gladness from the monumental *Roman de la Rose* (complete by c. 1270), who leads the central “*karole*” in the garden. “She had a voice that was both clear and pure; about her there was nothing rude, for she knew well the dance steps, and could keep good time the while she voiced her song” (Robbins 1962, 16).

That singers and dancers of *caroles* were one grounds inquiry into interplay between these paired kinds of movement through time. The close connection between music and dance is immediately apparent in the natural fit between binary dance steps, alternating left and right, and music in binary mensuration, imperfect *tempus*.⁹ Some sophisticated virelais of Guillaume de Machaut (1300-77) will be shown to layer grouping, tonal, and formal irregularities upon the steady metrical foundation of the *carole*’s origins. Throughout, I explore the question of dancers responding to certain types of metrical flexibility and ambiguity made possible through *ars nova* mensural innovation, a testable premise without definitive medieval answers. Specifically, how can dance interpret a stream of pulses that are weighted unequally by the feet and grouped irregularly in a variety of musical ways? The metrical and tonal tensions explored

⁸ Such is the performance practice described by Machaut in the *Remede de Fortune*, lines 3437-41; see Peraino 2011, 236. Mullally 1986, 226, also observes that a *carole* may be sung entirely by a soloist (refrain and verses).

⁹ That fourteenth-century dance songs should feature simple, consistent metrical groupings is taken as obvious by many modern scholars. Earp 1991, 124, for example, states that “we hardly need to mention that the rhythmic projection of such a song would have been strictly metrical, something to move the feet to.” Mullally 1986, 231, points out the mistake of transcribers who set this repertoire in 3/4 based upon its common trochaic rhythms (set as half-quarter). The binary pattern of the *carole*’s steps (L-R / L-R) suggest rather a binary beat system that can accommodate trochaic rhythms: 6/8 (quarter-eighth).

in my analyses of danced song reveal something that cannot be seen in manuscript or in modern score, nor can they be felt in a concerted performance today. Yet the danced embodiment of meter and its interplay with voiced song may have been top of mind to the medieval composer, singer, and dancer.

My theorizing about fourteenth-century music draws inspiration from later sources that more explicitly tie metrical grouping to physical movements such as the lowering and raising of the hand, recurring in two-part succession, an outward manifestation often related to the duple timekeeping of the heartbeat.¹⁰ According to Aristotelian doctrine, “time was only rendered sensible as a description or quantification of enacted motions or changes” (Grant 2014, 15). If conducting with the hand represented for Renaissance musicians a physical theorization of mensuration—a way of measuring and organizing notated durations, especially for the benefit of coordinated performance—I suggest that motion of the other limb, stepping with the feet, can equally accomplish the same in the repertoire of medieval danced song. To gloss Ornithoparchus’ *Musicae activae micrologus* of 1517: “It is the *motus*, signaled by the hand [foot] of the chief singer [dancer], which directs the measure of song in accordance with the (mensuration) signs” (quoted in Grant 2014, 31).

I will introduce my analytical methodology through a simple dance song from thirteenth-century France—catchy, short, and repetitive—before delving into the more complex virelais of Machaut in the following century. Also a composer of courtly motets and a mass, a number of Machaut’s virelais feature a metrical, melodic, and even contrapuntal

¹⁰ Grant 2014 interprets writings on meter from the sixteenth and seventeenth centuries, which “conceptualized meter through motion,” in dialogue with modern theories. On the *battuta* of Zarlino, see 5-6 and 33.

sophistication that may overflow the modest cup of round dancing.¹¹ Yet we have Machaut's word paired with illustrations in the manuscripts he oversaw that this particular genre of song was still danced in the round; at least he wanted them represented that way. Indeed, the composer consciously and persistently used the term *chansons baladées* ("danced songs"), perhaps in an effort to cling to the dance origins of what his peers and we today call the virelai.¹² In Machaut's words: "I began this virelai / which is called a danced song; / thus it should be proclaimed" (*Remede de Fortune* lines 3448-50, trans. in Peraino 2011, 236). Following Machaut, this study of dance as metrical analysis singles out the virelai amongst the other lyric genres.

In *De musica* (c. 1300), Machaut's compatriot Johannes de Grocheio coined the Latin term *ductia* for secular dance music (after *duco*, a leader dance). His basic descriptions of three types of vocal *ductia*—round, estampie (*stantipes*), and a species of dance music Grocheio also called the *ductia* (*per se*)—reveal the *ductia* (*per se*) to be the ancestor of virelai form danced in

¹¹ Earp 1991, 113-14, credits the infiltration of rhythmic innovations applied earlier to *ars nova* motets into secular song with its dissociation from dance. Earp 2005, 106, "believe[s] that the early virelais of Machaut are better understood as highly artistic works that merely evoke characteristics of the old dance genre."

¹² Peraino 2011, 236, points out the same terminological insistence in Machaut's *Prologue*: Section V, lines 14-15. She attributes Machaut's connection of the virelai with dance to French refrain songs called *rondets de carole* and *ballettes*, and to the Occitan *dansa* (248). According to Earp 1991, 115, "it appears that Machaut was quite self-consciously preserving this genre [virelai] for the dance." Referring to Machaut's virelai "Dame, a vous sans retollir," to be discussed below, Leach 2011, 159, notes that these songs are, for him, "very specifically a social dance form."

a *carole* in late thirteenth-century France.¹³ Similar to Grocheio's generic and specific usage of *ductia*, the association of caroling with the virelai alone seemed to have crystallized around 1300 out of a more generic sense of *carole* as a dance to be shaped according to the form of a song's lyrics.¹⁴ For this reason, thirteenth-century references to the music for caroling appear rather general: "*cançon de carole, chant de karoles*," among others (Mullally 1986, 226).

I begin with a characteristically straightforward *carole* from Jacquemart Gielée's romance *Renart le Nouvel* (1291-97). A modern transcription is given as **Example 2**.¹⁵



Example 2. transcription of a thirteenth-century *carole*

Sung by a fox, this love song is comprised of two parallel phrases, both essentially stepwise. The antecedent, mm. 1-3, is a series of tumbling cascades to an open (*ouvert*) cadence on E.¹⁶ The consequent, mm. 4-6, is exactly parallel, finishing a note sooner with a

¹³ See Chaganti 2018, 196, Mullally 1998, 10 and 17, Mullally 1986, 229, and McGee 1989, 499-506. Distinctions between Grocheio's three species of *ductia* are mentioned below.

¹⁴ McGee 1989, 509 n. 29. McGee's valuable study of Grocheio's survey of secular dance music at the turn of the century equates the "vocal *ductia*" with the carol (511).

¹⁵ Lightly adapted from Mullally 2011, 119, a transcription of F-Pn fr. 25566, f. 129r (refrain vdB 1797). In Mullally's translation: "My very sweet lively lady, listen to my heart beseeching you."

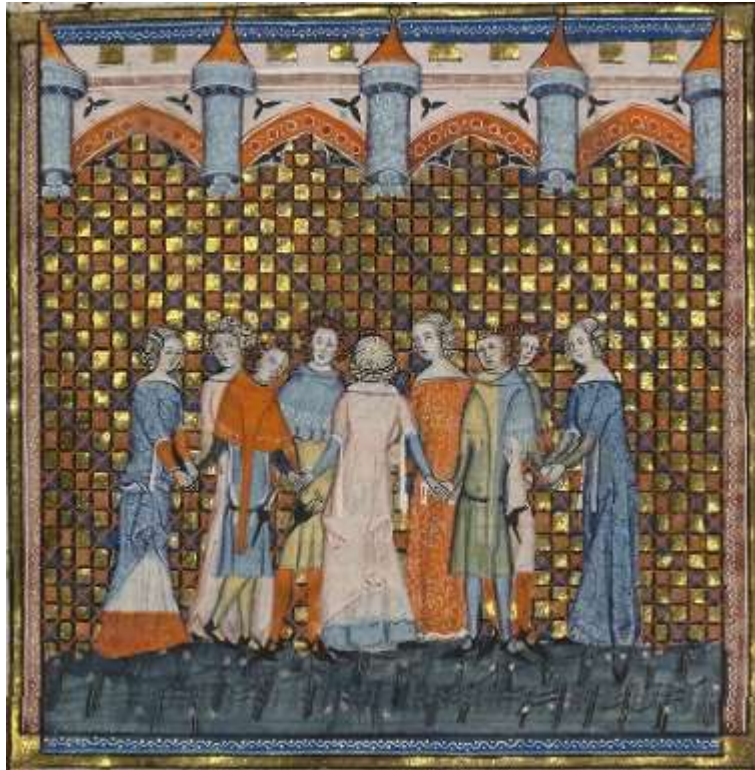
¹⁶ An open (*ouvert*) cadence is identified with reference to a song's terminal pitch and implies future resolution in a closed (*clos*) cadence. Scale degrees ^2, ^5, and ^7 are customary sites of open cadence, all ultimately seeking ^1.

closed (*clos*) cadence on F. Such music is aptly described by Grocheio, for whom the *carole* [*ductia*] is “a light dance song, rapid in its ascent and descent” (McGee 1989, 506) and which pairs open/closed (*apertum/clausum*) endings. Grocheio contrasts the lighter *carole* with the *estampie*, which is both longer and more difficult, lacking regular metrical stress.¹⁷

The fourteenth-century *carole* need not be any more complicated. A 1338-44 manuscript of the *Roman d’Alexandre* provides music for the rondeau “Ensi va qui amours” alongside sumptuous illustrations of people singing and dancing in the round, as that of **Example 3**.¹⁸ My transcription of the song in **Example 4** is underlaid note for note from the same page of manuscript.

¹⁷ Mullally 1998, 22, and McGee 1989, 509-10.

¹⁸ Both Examples 3 and 4 come from MS Bodley 264, f. 181v. While “Ensi va qui amours” is the only rondeau of this study otherwise concerned with virelais, the interpretation of grouping patterns through alternating dance steps is abstract enough for pertinence to the melodic styles of both lyric genres. Moreover, the earliest *caroles* were associated with lyrics in rondeau form in the *Roman de Guillaume de Dole* (c. 1228); see Mullally 1986, 226. Peraino 2011, 249, reviews the lack of distinction between rondeau and virelai before Machaut. Writing c. 1300, Grocheio observes that the rondeau sets refrain and verse to the same music, as here, while the other vocal dances, *ductia* (ancestor of the virelai) and *estampie*, set verses to differing music and include pairs of open/closed endings (McGee 1989, 499-500). Chaganti 2018, 196, notes that “song in *virelai* form came to replace the rondeau as an accompaniment to *carole*.”



Example 3. *carole* dancing to the rondeau “Ensi va qui amours”

Ensi va qui amours demaine a son command

En-si va qui a - mours de - mai - ne_a son com - mant

Example 4. manuscript and transcription of “Ensi va qui amours;” refrain

The melody again tumbles downwards in each of two parallel phrases, essentially arpeggiating through a triad: C down to A in mm. 1-2, A to F in mm. 3-4. **Example 5** proposes

dance steps for “Ensi va qui” following two axioms.¹⁹ First, the dance should start with the left foot so that it will naturally revolve clockwise. Second, the dancers should mark time by stepping on the beat, defined here as the breve; that is, the steps articulate the *tempus* level of mensuration. The beat level may also be considered the long or semibreve depending upon the notation level and preferred tempo, how quickly the dancers want to revolve.²⁰



Example 5. Refrain of “Ensi va qui amours;” marked for dance

¹⁹ Inspiration for this simple dance notation was taken from Little 1975.

²⁰ Concerning notational options in the *Ars nova*, Jacques de Liège (1320s) observes that “the perfect breve has the same value as had the perfect long for the Ancients,” with consequences for modern transcription; see Maw 2004, 47. The rates at which to sing and to dance a *carole* are interrelated yet can be adjusted separately to expressive effect. One can choose to step fast (*tempus* level) or slow (*modus*) without changing the sung tempo. Doubling or halving the music sung per step in this way produces quite a different feel in performance and can impart a livelier or more somber expression, as desired. Changes to sung tempo increasingly call for changes to dance pacing in inverse proportion. As a song’s tempo is quickened, there is increased pressure to dance it at the next broader mensural level lest the step break into a jog. On the contrary, if a song is taken slowly, the dancers may better subdivide to avoid dragging their feet.

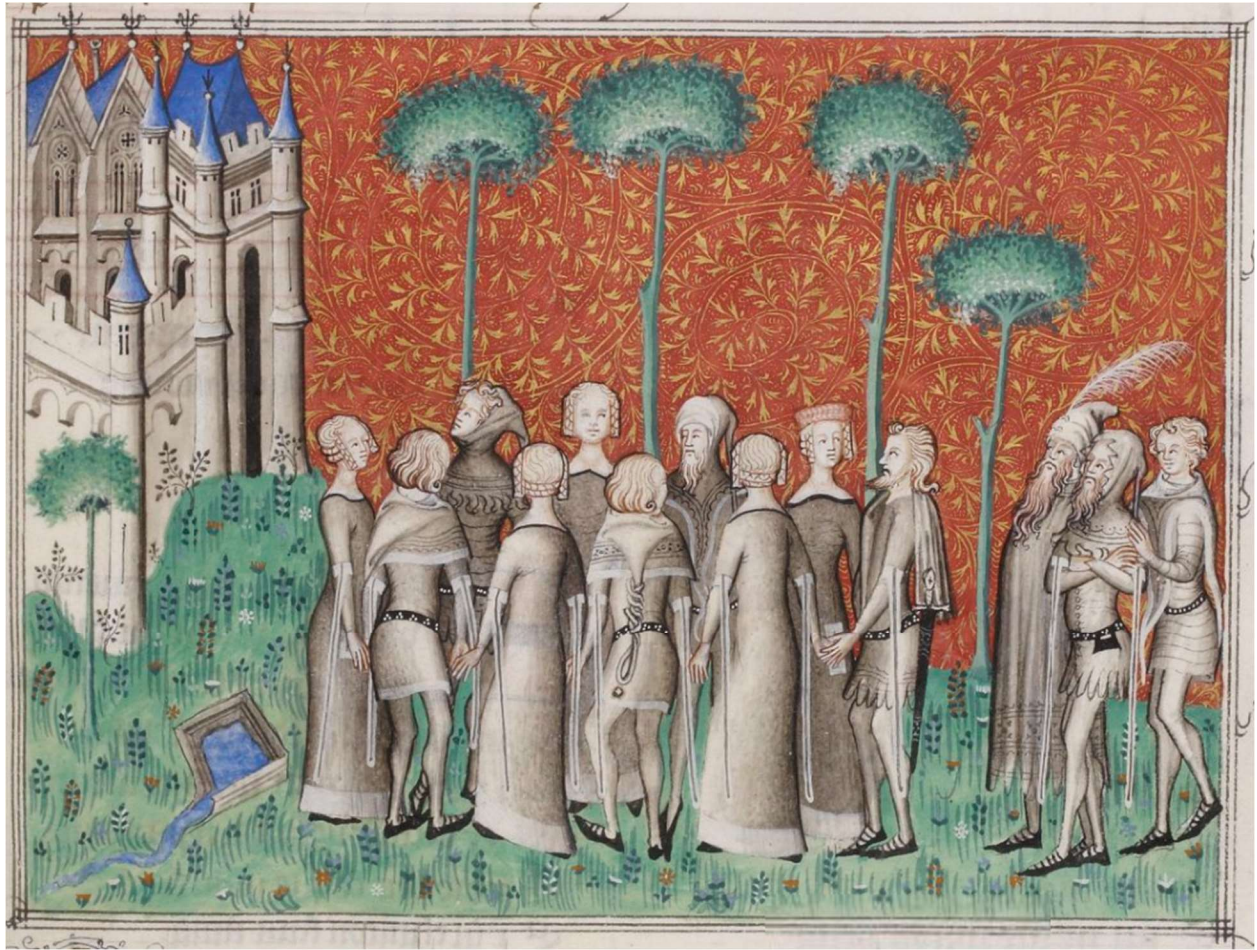
This step grouping—little more than walking—is unproblematic and wraps back onto itself across repeats.²¹ The danced songs of Machaut are of another order. I will first consider the relatively simple danced song “Dame, a vous sans retollir,” from Machaut’s *Remede de Fortune* (c. 1340s).²² While the musical interpolations of the *Remede* include all the *formes fixes*, Machaut associated dancing with this *chanson baladée* by setting an illustration of danced performance beside the score of this song alone, given as **Example 6**.²³ Here, carolers sing and dance a *virelai* led by its composer.²⁴

²¹ Notice how the musical setting of a poem can transform its natural pattern of declamation. The poetry of these two 6-syllable lines is iambic (scansion marked), yet the musical rhythm almost entirely reverses the natural stress of the text. In that context, the coincidence of poetic, metrical, and agogic stress on “-mours” and “-mant” marks the end of each line. The analyses to follow explore how poetic and musical rhythms can suggest alternative metrical groupings. Musical rhythms, as potentially transformative of the former, are weighed more heavily.

²² See Peraino 2011, 239-42, for detailed elucidation of this song’s lyrics and function within the *Remede*.

²³ Earp 1991, 126, claims that the story line of the *Remede* was contrived to demonstrate each type of song then current. See also Earp 2005, 105-6, on the *virelai* as danced song in the *Remede*. Peraino 2011, 235, calls *Remede* “in effect a musical *summa*.” See Butterfield 2002 on the generic characteristics of *romans à chansons* and Stokes 2014 on the didactic undertones of Machaut’s *Remede*, particularly regarding medieval rhetoric. Leach 2011, 139-64, provides an excellent summary of the *Remede* and analysis of its lyric interpolations in the context of the poetic functions of Hope and Desire.

²⁴ According to Peraino 2011, 285, “there we see the singer Machaut hand in hand with women in the dance. His voice in the *virelai* enables the carole.” See also Leach 2011, 159, on Machaut’s identity in this song and its depiction.



Example 6. dancing Machaut's virelai "Dame, a vous sans retollir;" detail from *F-Pn* fr. 1586

(Mach C), 51r

A modern transcription of the song follows as **Example 7**, including dance steps on every beat following the axioms above and formal labels A and B.²⁵ The A section (refrain) opens with

²⁵ My transcriptions of Machaut are based upon Ludwig 1926, differing sometimes in suggested ficta and bar line placement, unless otherwise noted. In this case, however, I prefer the anacrustic transcription of Maw 2002, 82, for its benefit of situating both cadences on downbeats, contra Ludwig and Schrade. My dance steps L and R in "Dame, a vous sans retollir" correspond to what Earp 1991, 129, calls "metrical units" 1 and 2, adjusted for the anacrusis. Earp 2005, 106-7, uses the same song as an example of conventional text declamation. His bold-numbered syllables there correspond to my suggested dance steps.

tonal articulation of G (mm. 2 and 5), which become subordinate to the final cadence down a step on F (m. 11). The B section floats around C, an open $\wedge^5$ in reference to the final.

A R L R L R L R L R L R L R

8 Da-me, a vous sans re-tol - lir | Dong cuer, pen-se-e, de- sir, | Corps, et a - mour, | Com-me a tou - te la mil-
Vos - tre biau - te fait ta - rir | Toute autre et a - ni-en - tir, | Et vo dou - cour | Pas - se tout; rose en cou-

7 L R L R L R L R L

8 lour | Qu'on puist choi - sir, | Ne qui viv - re ne mo - rir | Puist a ce jour.
- lour | Vous doi te - nir, | Et vo re - gars puet ga - rir | Tou - te do - lour. |

B 11.5 R L R L R L R L R L

8 Si ne me doit a fo - lour | Tour-ner, se je vous a - our, | Car sans men-tir,
Bon-te pas - ses en va - lour, | Tou - te flour en douce o - dour | Qu'on puet sen-tir. |

Example 7. Machaut's "Dame, a vous sans retollir;" marked for dance

Compared to the *Renart* chanson a generation prior (Example 2), Machaut's dance song is longer, traverses a wider range, dabbles in chromaticism, and freely mixes trochaic and iambic subdivisions.²⁶ In whole it demands more from an amateur chorus of singing dancers. Yet because it lacks any grouping disturbances at the beat level, to dance this song merely adds locomotion to tapping along with one's foot. In the anacrustic transcription here, the song's cadential layout is representative in tying cadential arrivals and text line endings to the left foot; ending both the A and B sections and marking every downbeat. This is a strong and

²⁶ For Earp 1991, 138, "Machaut's early dance virelais... are already far more refined than a popular dance song."

Peraino 2011, 356, finds the relatively "off-kilter rhythmic environment" of Machaut's virelais to result in "a lively tune that dances itself, though it may not be itself fit for dancing."

characteristic correlation in medieval dance music. Machaut's virelais cadence on the left foot 91% of the time out of about 120 cadences.²⁷

The generic linking of a particular step to tonal articulations provides an illuminating tool for analysis. For example, consider the unusual conclusion of the B section (m. 11) in Machaut's danced song "Loyaute weil tous jours," given as **Example 8**.

8 L R L R L R L R L R L R L R

Loy - au - te weil tous jours mein-te - nir et de cuer ser - vir ma da-me de - bon - nair - re.
 Son tres doulz vo - loir sans re-pen-tir | Et li o - be - ir Com a mis, sans m'ef - fai - re. |

8 L R L R L R L R

Mon cuer y weil et mon de-sir met - tre sans re - trai - re.
 Ne ja ne m'en quier de-par-tir, Eins weil tou-dis fai - re |

Example 8. Machaut's "Loyaute weil tous jours" (V2); marked for dance

The pitch center of this song is F, confirmed especially by the refrain's final cadence (m. 7). The unequal line lengths of the poetry are distributed by the melody to end mainly on tonal pillars F and C, once touching on A (m. 5) on the descent through the Lydian pentachord. Dancing three times through the refrain of this virelai establishes F (^1) as a left foot kind of note, and it is therefore unusual for the B section to end with F on the right (m. 11). The

²⁷ Cadence here refers to arrival on a pitch stressed by alignment with the end of a poetic line, agogic accent, being followed by rest, modal relationship to the last note of the song, and by perfect consonance with other voices in polyphonic cases. Moll 1998, 32, provides a useful table for defining cadence in fourteenth-century French masses.

regular step groupings L—R / L—R provide a medium through which to feel the weakness of the melody's hop down to F and excuse it. This right-footed F is further attenuated by the natural declamation of the poetry's feminine ending, where the penultimate syllable of "re-TRAI-re" is stressed (left foot) and the final syllable unstressed (right). In a typical *forme fixe*, the B section would terminate on an *ouvert* scale degree, not the *clos* final of the song. So what looks formally astylistic in "Loyaute weil tous jours"—a verse ending on the final—does not feel like it because ^1 is relegated to the conventionally weak right foot. In other words, a pitch that is melodically *clos* is here felt as metrically *ouvert*, as expressed by the dance.

This analysis hints at an important implication arising from the norm of an accented step in medieval dance music: meter. Meter is not explicitly indicated by fourteenth-century French mensuration, which was a quantitative system of notating discrete lengths of pitch and rest.²⁸ Rather, meter is a qualitative interpretation that accrues from a regular, hierarchical grouping of pulses. Once established, a metrical grouping can be felt to remain in effect until its pattern of accentuation is contradicted by some alternative. It is possible, as will be seen below, for entirely regular mensural notation to be interpreted (and danced) in irregular metrical

²⁸ Boone 2000 undertakes a detailed investigation of regular rhythmic punctuation in the layered mensural pulses of Machaut and others across the medieval and Renaissance periods. Boone then contrasts mensuration as temporal framework with common practice meter. Succinctly put by Maw 2002, 70, "while the theory of mensuration supplied a quantitative understanding of musical time sufficient to make sense of rhythm for the purposes of notation, metrical thought provided a supplementary qualitative understanding of musical time necessary in practice to make sense of rhythm for the purposes of musical effect."

groupings if the musical organization suggests it.²⁹ Following David Maw’s extensive work on meter in Machaut’s songs, I identify the sense of medieval meter as a combination of motivic initiation and termination, rhythmic grouping (especially recurrent), agogic accent, text alignment, simultaneous articulations, contrapuntal resolution, and harmonic change.³⁰ Indeed, “metrical organization is a necessity for dance music, as the coordination of bodily movements requires audible marking of the bar in the construction of the music” (Maw 2004, 61). In dance music overwhelmingly notated in duple mensuration, meter is embodied and entrained as the regular grouping of left/right steps accrues a sense of strong/weak, especially because the same foot aligns with tonal accent in the vast majority of cases. The way that alternation of left/right feet in dance parses mensural groupings into strong/weak maps onto the hierarchical organization of medieval French mensural theory insofar as any subdivision will have its first attack accented by simultaneity with its parent duration.

Having established some musical conventions of *caroling* above, I endeavor to address Machaut’s *chansons baladées* that map onto the dance less easily.³¹ The question that I am

²⁹ Examples of the flexible metrical interpretation of regular mensural notation can be found throughout Maw 2004 and 2013, including transcriptions in mixed meters.

³⁰ Maw 2004, 62 and *passim*, addresses the *modus* barring methodologies and inconsistencies in the complete editions of Machaut’s secular songs by Ludwig and Schrade. See Maw 2002 for a systematic investigation of *modus* in Machaut’s virelais, a more flexible mensural level that requires analytical interpretation of musical groupings, supplemented by a focus on text setting and meter in Maw 2013. A complementary analytical account of flexible *modus* barring in Machaut’s motets is undertaken in Lavacek 2022.

³¹ As noted in Leech-Wilkinson 1991, 16, “hardly any have the regular metrical structures one would expect of a dance accompaniment.”

concerned with here is not so much whether these songs of Machaut were danced in his century, but rather how a dance pattern of simple alternation can foreground the increasing metrical flexibility of the *ars nova* at various mensural levels. As with the previous dance songs, Machaut's virelai "J'aim sans penser laidure" can be transcribed in 6/8 and alternating steps assigned to each beat like in **Example 9**.

Example 9. Machaut's V14, "J'aim sans penser laidure" (refrain only); marked for dance

In this execution of the dance, the usual left foot downbeat is in force until m. 8, where a relatively long stream of eighth notes, free from dependence upon a neighboring quarter to capture them as trochees or iambs, provides an opportunity to consider how a change in grouping—expressed in dance steps—can shape a narrative of instability to stability across a refrain. While a change in mensuration is not represented in Machaut's notation, Example 9 follows the edition of Leguy in rendering a text setting of one syllable on each of three equal beats in m. 8 and again, the same motive with textual assonance, in m. 12.³² The tight spacing

³² Following Earp 1991, 130, "there are no extended melismas, and thus the ear hears the text syllables themselves as providing the musical rhythms." The syncopated "pitch accents" marked in Example 1 of Maw 2002, 72, occasionally insinuate a perfect minor grouping that rubs against the prevailing imperfect major metrical and

of the consecutive eighth notes in mm. 8 and 12 followed by a generous gap before the final quarter note—demarcating beat 3 of 3/4 rather than breaking up an iamb of 6/8—in all manuscript sources can be suggestive of a grouping change.³³ But the spacing, the system breaks (Mach C), and even the page breaks (Mach A) that occur in some manuscripts between the last F and A of mm. 8 and 12 more likely reflect the line break in the poem (indicated by vertical stroke in the examples), inked in before the music notation was added. However, the way that Machaut repeats the structural pitch A in mm. 7-8, as well as details of text declamation, *can* be used to define logical metric groupings.³⁴ In V14's melody, the final D (^1) is decorated with lower and upper neighbors before descending through the Dorian tetrachord D-A, closing on ^5 in an *ouvert* cadence (m. 5). In m. 7, A marks the onset of each of two beats (in 6/8), while in m. 8 the same A can mark each of three beats (a change to 3/4); the same duration segmented differently. The word "*na-TU-re*" ends the third line of the poem. In Machaut's rhythmic setting, "*na-*" is set as a pickup note to the stressed "*-TU-*" on the

declamatory accents of Machaut's V1, "*He! Dame de vaillance.*" Example 14c of Maw 2002, 91, explicitly renotates V18, "*Helas! et comment,*" along the same lines, switching to 3/4 to capture a measure of rhythm identical to that of mm. 8 and 12 of V14 as I have rendered them in Example 9. Example 8 of Maw 2013, 397, presents essentially the same transcription of V18 in 6/8 (mm. 1-6) then 3/4 (mm. 7-11) based upon the syllabification of those phrases. Peraino 2011, 242, mentions a similar "metrical ambiguity (toggling between 6/8 and 3/4)" in Machaut's "*Dame, a vous sans retollir,*" examined above, without further specification.

³³ A subtle separation of the first three minims from the fourth in these measures (A-G-A then F) in all manuscripts likely suggested the continuation of 6/8 time to Ludwig and Schrade, contrary to the metrical flexibility pursued in my Example 9 and the edition of Leguy 1977.

³⁴ Earp 2005, and its 2012 extension, document the conventional relationship between metrical pulses and rate of text declamation in Machaut, regarding irregularity ("declamatory dissonance") as a strategy of poetic expression.

metrically accented beat 1 (m. 8). The weak syllable “-re” follows naturally on a weak beat—beat 2 in both my 3/4 reading and in Ludwig and Schrade’s 6/8 reading, but aligned one eighth note later (as suggested by the manuscript spacing)—and this could have ended the measure. Yet Machaut sets another weak syllable, “ont,” in m. 8 (beginning the fourth line of the poem) in a way that acts as a pickup into m. 9, descending now through the Dorian pentachord A-D.³⁵ Mm. 11-12 are slightly different. F marks the two beats of m. 11 before A returns to mark the three beats of m. 12, where again the final A is detached as a pickup to the final descent A-D. Such is the grouping found in Leguy 1977 (from Mach G), an edition remarkable indeed for its flexible approach to metric grouping as a response to Ludwig’s supposed rigidity.³⁶ My **Example 9** represents an exploration of the ramifications of flexible grouping, a trend gaining modern currency. David Maw’s recent work (2013, 2004, 2002) explores a similar responsiveness to irregular metrical/textual groupings in Machaut’s songs, and his transcriptions of some *chansons baladées* also feature rebarrings and changes of time signature. On the contrary, Ludwig and Schrade beam mm. 8 and 12 as continued iambs in 6/8.

My reconfiguring of dance steps in these measures and in the examples to follow is an act of modern metrical analysis and makes no claims about medieval practice. The dance steps

³⁵ This Dorian pentachordal descent is balanced either by the Dorian tetrachordal descent D-A described above (mm. 1-5) or by the pentachordal descent E-A defined by the poetic line breaks, in the latter case linking two locations of Dorian pentachord.

³⁶ Along with V14, Leguy 1977 assigns the dual time signature 6/8 and 3/4 to virelais 2, 5, 8, 16, 19, 26/23, 27/24, 31/28, and 32/29 (numbered as in Ludwig/Schrade), among others with multiple time signatures, and freely beams groups of two and three eighth notes in order to capture different “rhythmic cells” (see Foreword, VI).

above the staff in m. 8ff convey a perceived change in metrical grouping, swapping imperfect time/major prolation (compound duple) for perfect time/minor prolation (simple triple).³⁷ The opposing groupings 2x3 and 3x2 are of equivalent duration—a demand medieval mensural theorists called equipollence—but where should the dancers’ feet go? Suppose that stepping with the beat is so conventional that dancers will do so even when the beat changes, as shown in **Example 9**. The consequences of taking triple time in m. 8 ripple throughout the next few measures (in gray). The time in mm. 9-11 is duple as usual but the accent has been temporarily displaced to the right foot, until a second grouping change of the same sort (m. 12) realigns the left foot with the metrical stress (in black). The drama of an occasional metrical disturbance in these songs resonates with Machaut’s courtly love poetry in the *chansons baladées* genre. Singers of virelais often professes themselves to be at great pains to maintain steadfast loyalty to a beloved paragon, whatever may come. Multimedia performance—combining lyric poetry, song, and dance—allows for higher level meanings to emerge as one medium comments upon the other. Reevaluating the genre through the lens of dance, as I advocate, grouping disruptions can impart a cross-footed tension that is impossible to detect in manuscript, in modern editions, or in concerted performances.

A brief but constructive comparison may be made here between fourteenth- and sixteenth-century conceptions of triple grouping. Some Renaissance theorists witnessed an

³⁷ Time (*tempus*) is defined by the triple or duple (perfect or imperfect) subdivision of breves by semibreves. In most modern transcriptions of fourteenth-century French music, as here, the breve is represented by a measure and the semibreve beat by a quarter note (dotted when perfect). Prolation (*prolatio*) is the triple or duple (major or minor) subdivision of semibreves by minims, transcribed here as eighths per (dotted) quarter.

irregular parsing of triple time, one rooted in the primacy of binary groupings as regulated by the conducting hand and derived from the heartbeat, as mentioned above. Whereas in my methodology every pulse (at a given level) is marked by a step, in this sixteenth-century conception, the first of two beats is elongated to twice the length of the second, yielding an unequal duple 2:1 proportion comprised of three pulses.³⁸ To extend this motion to the feet in **Example 9** is to maintain the usual L/R pairing throughout mm. 8-12, not with equal duration L R | L R as in the foregoing imperfect *tempus*, but with an extended lurch of the left foot: L— R | L— R. An interpretation of unequal duple here can be supported by the same reasoning that concerned distinguishing a beat 3 of equal triple above, the considerable conceptual difference between them being expressed by the consistency, or not, of the left foot marking downbeats and in how the internal grouping of each measure is felt as anomalous: either by a quick pitter-patter of three steps or by a lurching of two. The theoretical underpinnings of these choices are significantly different. However widespread stretching the downbeat may have been in a sixteenth-century metrics rooted in duple alternation, *ars nova* mensuration rather had the triple “perfections” of the *ars antiqua* as its foundation, of which duple was considered an imperfection.³⁹ I find no fourteenth century reference to dancers elongating a single step to cover two semibreves as a means to preserve a norm of binary groupings. However a triple

³⁸ Grant 2014, especially chapter 3, brings this historical kind of triple meter into dialogue with modern theories using examples from the sixteenth century through Handel. The consideration of medieval music here responds to Grant’s call to “envision yet other pieces under the same distorted lens: to see the inequality hiding just behind the surface of the music” (78).

³⁹ For example, a breve may be imperfected by an adjacent semibreve, reducing the former to 2/3 of its original value. Thus is produced a 2+1 triple group by medieval subtraction (from 3), rather than Renaissance addition.

group was to be negotiated, in our century or theirs, it falls to a song's leader to demonstrate its steps no less than to demonstrate its pitches and words.

Let us return to the analysis of Machaut's danced songs through the framework of the *carole*. Metrical disruptions like considered above in V14 can also be found in Virelai 27/24 (Ludwig/Schrade), "En ma cuer," only reversed. In that song, the prevailing meter is triple and the disruptions duple.⁴⁰ In **Example 10**, striking pairs of iambs on the surface are notated in 6/8 on a separate staff above the score set in 3/4. In the iambs' first instance (m. 3), the accompanying tenor reinforces the cantus' lilt for a single measure. Because an iambic reading yields only two steps per measure, the established dance pattern becomes offset (in gray above, compare to black below). A second change of this same type (m. 7) sets the steps back on track in the cantus alone, the tenor remaining in perfect time.⁴¹ There are five such measures of ambiguous groupings and if they are consistently taken the song's footing will end opposite to how it would have if taken straight. This is not necessarily a problem, however, because dancing the occasional duple measure sets a few directed progressions onto left foot arrivals when they would otherwise land on the right, including at the end of the song. That is, being swayed by the iambs into a metrically flexible approach to this song marks the agogic and tonal stress of mm. 4 and 22 on the final F with a conventional left foot while subordinating

⁴⁰ Fuller consideration of dancing those rare virelais set in triple time is taken up below.

⁴¹ A disjunction between two grouping patterns in a polyphonic song begs the question of which grouping the dancers followed, assuming they all stepped together, and likely excludes (complex) polyphony from being danced. As argued throughout this study, and in much work by Earp and Maw, Machaut's virelais as a genre still bear the imprint of dance however artful they became.

directed progressions to C (m. 8) and G (m. 14), common scale degrees for open endings, to right foot status.⁴²

Example 10. Machaut's V27/24, "En ma cuer" (refrain only); marked with two options for dance

We revisit the binary Virelai 14 now to consider how dance may illuminate other metrical issues in this repertoire. The first sung pitch would involve the right foot if any step is to be taken on this beat.⁴³ The opening dotted quarter rest of **Example 9** is a transcription of

⁴² These are highlights of a larger tonal arch fashioned by the poetry's line lengths. Set in groups of 7 and 4 syllables, as marked with vertical strokes in the example, the lyric endings define an orderly plan initially rising in fifths F-C-G (mm. 4, 8, 10), pausing on G (m. 14) as an open neighbor to final F, then descending back through fifths G-C-F to close (mm. 16, 20, 22).

⁴³ Little 1975, 114 and 119, provides a French Baroque solution to the issue of pickup notes in danced music, though it cannot be securely backdated to fourteenth-century practice. One of the first European dance manuals

the semibreve rest found in all sources (Mach A, C, G, and Vg); omitted by Ludwig 1926 and restored by Schrade 1956.⁴⁴ While an opening rest may make little difference to singing alone, its semibreve duration joins with the first note to complete the characteristic mensural unit of imperfect breves in danced songs and, of special interest here, a rest provides the necessary space for dancers to begin stepping clockwise. Recall that in dancing the *carole*, the left foot advances on downbeats while the right snaps back to it on upbeats. **Example 11** shows this song's opening in the clear notation of Mach C. It is even more remarkable that a breve rest precedes the semibreve rest just discussed, the breve representing a full measure in transcription (noted but omitted by Ludwig and by my Example 9 because it does not affect dance steps at the *tempus* level). Because these opening rests appear consistently in all sources, the empty breve (measure) would appear to serve an important function in the *modus* organization of this song. If m. 1 of my transcription is taken as a hypermetrical upbeat, adopting the empty breve as the downbeat of m. 0 (not notated), then all formal sections of V14 cadence more emphatically on *modus* downbeats (mm. 14, 18, and 20 of Ludwig). If m. 1 is felt to start the song on a *modus* downbeat, on the other hand, all formal boundaries—indeed, the end of the song—close on a weak beat at that level. While a rest of a measure's duration is ripe for signification, four of Machaut's thirty-eight *chansons baladées* (V5, 9, 28/25, and 36/30) begin with a single eighth note pickup in Ludwig's edition. These are brief enough, I suggest, to

to be inlaid with corresponding musical notation shows pickups being danced with a plié before the steps proper began.

⁴⁴ With V14, V4, 27/24, and 36/30 also begin with a semibreve beat of rest in all manuscripts. Ludwig reproduces only that of V27/24, likely to counterpoint the tenor's entrance, while Schrade begins both V27/24 and 36/30 with a beat of rest (both polyphonic) but not V4 (monophonic).

be sung before any dancing commences with the left foot on the ensuing downbeat.⁴⁵ An awareness of dance explains why measured silences may have been worth the trouble of notating for a medieval composer and audience. Opening rests were crucial not only for mensural reckoning but for performance insofar as they marked time so that the dance may start conventionally on a strong left foot before singing commenced on a weak right, a special feature of only about 10% of Machaut's danced songs.



Example 11. a breve and semibreve rest begin the dancing before the singing of Machaut's V14;

excerpt of *F-Pn* fr. 1586, f. 153v

The combination of song and dance in a *chanson baladée* may start in any of three ways, then. Either a given work starts both on the downbeat of m. 1, or it provides an opening beat of rest upon which to start the dancing before the singing, or it provides a short pickup note for

⁴⁵ In all manuscripts, V5, 9, and 28/25 begin with one or two minim rests (in imperfect and perfect time, respectively), enough to complete a full semibreve beat together with their single minim pickup note (excepting Mach E's rendering of V28/25). V36/30 begins with one semibreve rest and then a minim rest in all sources (Mach A, G, and Vg); that is, a full beat of rest followed by enough rest to complete a second beat of imperfect time with the single pickup note. Ludwig represents none of these rests while Schrade restores them all.

singing before a substantive downbeat for dancing in the next measure. In any case, I presume of fourteenth-century French *carole* practice that the assembled dancers looked to a given song's leader to commence a first step as appropriate, pickup or not, just as they listened for their first pitch before the group could catch on with singing and dancing.

The potential for metrical play in Machaut's *chansons baladées* certainly complicate if not rule out *carole* dancing. Maw, for example, finds the irregular metrical schemes of Machaut's virelais "unsuitable for dancing except of a stylized nature" (Maw 2002, 71). John Lydgate (c. 1426) may have associated such metrical irregularity with the vicissitudes of life's march towards death in his metaphorical poem on the *danse macabre*. Note especially how the frequent changes in footing are associated with varied measure in the words of Lydgate's minstrel (quoted in Chaganti 2018, 181).

This newe daunce / is to me so straunge
Wondir dyuerse / and passyngli contrarie
The dredful fotyng / doth so ofte change
And the mesures / so ofte sithes varie
Wiche nowe to me is nothing necessarie
If it were so that I myght asterte,
But many a man (if I shal not tarie)
Ofte daunceth but nothing of herte (lines 505-12).⁴⁶

Even if Machaut's *chansons baladées* overstepped their dance origins, the *carole's* pattern of alternating steps is pursued here nevertheless as an analytical tool extendable as far it can shed light on medieval metrical groupings. Even if practical dance is excluded, Seeta Chaganti makes the case that social dancing so permeated medieval French society that lyrics

⁴⁶ The complete work is edited in Cook and Strakhov 2019, see esp. 37.

like Machaut's were both crafted and read through the virtual embodiment of dance (Chaganti 2018, 2-4 and *passim*).⁴⁷ The virtual embodiment of dance is the sense in which poets took on the norms and limitations of dance music as a scaffolding for the creation of new lyrics in that tradition. Judith Peraino similarly claims that whether a lyric was actually sung or just read "has less to do with a change in the formal properties of the poetry than with a change in concept, namely, that a lyrical poem with a refrain and an origin in public dance songs could be a complete expression as read—without musical performance" (Peraino 2011, 259). Whether a given work was destined to be danced actually or not, its mensural grouping structure bears the footprints of dance and dance can (in any era) make that manifest. Either way, we gain a richer sensitivity to metrical grouping in Machaut's *virelais* whether experienced bodily or sung with an inner awareness (a virtual embodiment) of the dance.

While my reading of V14 featured a deviation in its grouping pattern—the intrusion of a triple group within a duple context—the song self-corrected via a second change that sets the whole right again. Other of Machaut's *chansons baladées* do not so provide for their own

⁴⁷ Chaganti 2018, 17, is convincing that "Machaut and Deschamps wrote lyrics that were mindful of the aesthetics of dance and dance accompaniment." The concern for musical embodiment, virtual or actual, that shapes my approach to fourteenth-century song here was part and parcel of musical interpretation in later centuries, too. Grant 2014, 16 (see also 27-9 and *passim*) concurs that "whether it was performed or served as a figure of imagined demonstration, the motion of the beat rendered temporal division sensible in these theories" of the sixteenth and seventeenth centuries.

metrical maintenance. Consider Machaut’s Virelai 5, “Comment qu’a moy,” annotated in

Example 12.⁴⁸

A (R) L R L R [1 2 3] L R L R L R L R L

Com-ment qu'a moy lon - tein - ne | soi - es, da-me d'on-nour, | si m'e-stes vous pro - chein - ne | par

7 R L R L B 1. R L R L R L R L R L R L R L R L R L

pen-ser nuit et jour. | Car Sou-ve-nir me mein-ne, | si qu'a des sans se - jour |

Example 12. Machaut’s V5, “Comment qu’a moy;” marked for dance

The refrain traces a stepwise path up from tonal center G to an open cadence on D (^5) in m. 4, then back down again for a closed cadence on G in m. 8. Both phrases set text of paired feminine-masculine endings that subdivide them at logical midpoints (see vertical strokes in the example). The first phrase is subdivided at B^b (m. 2, “lon-TEIN-ne”), halfway up to D. The second phrase recontextualizes the stability of the first’s masculine ending on D in m. 4 (“d’on-NOUR”) by setting a feminine one on the same D in m. 6 (“pro-CHEIN-ne”), figuring now in subordination to the coming final G. The B section floats around high G before settling down on the fifth, D, for an open cadence (m. 13).

⁴⁸ My example follows Schrade in representing the two minim rests that open both A and B sections in all sources of V5 (Mach A, C, E, G, and Vg), omitted by Ludwig. I follow both Ludwig and Schrade in taking the rest that follows the final G closing the refrain (m. 8) as a section break and not as a measured duration. This rest variously appears as a breve (Mach C, E, Vg) or semibreve with additional section break (Mach A, G).

Measured in the usual imperfect time/major prolation, I interpret a swap to perfect time/minor prolation in m. 3, transcribed explicitly with beaming and dual time signature (6/8 and 3/4) in Leguy's metrically flexible edition. Such equivocal time signatures can capture not only perceived changes in metrical grouping from one measure to the next, but simultaneous layers of grouping composed by Machaut, lyric and melodic, within the same measure. My consideration of grouping changes in V5 and elsewhere rests upon Machaut's transformative musical setting of his poetry, as melodic passages reshape the natural meter of the lyrics.⁴⁹ The typical iambic pattern of this six-syllable line (mm. 2-4) is "soi-ES, da-ME d'on-NOUR." In m. 3 of Example 12, the arrival of the accented syllable in "da-ME" could demarcate the second beat of 6/8. I argue, however, that when Machaut set this lyric to music the tonal articulation of the melody subtly recontextualizes the natural declamation pattern of its text, resulting in a change of perceived metrical grouping.

As mentioned above, the first tonal arrival in V5 is the D of m. 4. The ascent up from G reaches this D earlier, on the downbeat of m. 3, but does not come to rest as in m. 4. Instead, the premature D on beat 1 of m. 3 turns on a neighbor C on the weak part of the beat. The same D is touched again on beat 2 (in triple time) before skipping down to B^b. This B^b does not have the force of the downbeat B^bs in mm. 1 or 2, nor do I hear it to initiate beat 2 of 6/8 time. This B^b rather decorates the cadential line D-D-C-D across mm. 3-4, all on the beat. While m. 1 has the same syllabic rhythm as m. 3 (trochee followed by iamb), melodic design distinguishes their metrical grouping. The repeated Gs of m. 1 divide the measure in half. The first melodic

⁴⁹ I thank my anonymous reader for drawing my attention to the importance of line length and declamation patterns as aspects of Machaut's multilayered grouping structure.

contour descends a third from B^b to G, with passing tone A between them. The second contour departs from the following G on beat 2 (“*moy*”) down to F. This F acts as the local goal tone of m. 1 and not a lower neighbor to G (potentially *on* the beat, if this measure were grouped in 3/4) insofar as the ensuing leap of a fourth—F to B^b in m. 2, as wide a gulf as the song contains—portrays the distance that separates the protagonist from the noble lady. The single metrical disruption pursued in m. 3 of Example 12, then, results in a refrain that ends off-kilter.

The melodic shape of the B section suggests another triple group in m. 9, similarly reinterpreting the grouping pattern of the iambic lyrics. Where the typical declamation of “SOU-ve-NIR” accentuates beats one and two of 6/8, the melody’s repeated Gs of mm. 9-10, parallel to its repeated Ds in mm. 3-4, can set off distinct beats. Opening on a pickup E, m. 9 steps back down G-F(♯)-E across three beats, as shown. This third leap is then outdone by the leap of a fourth in m. 10 as remembrance of the lady guides the protagonist (“*Car Souvenir me meinne*”) across the same wide interval that separated them in the refrain. The B section concludes with an *ouvert* ⁵⁰ (D) the first time through, then more solidly on the left when that section repeats (in the standard formal representation AbbaA), which sets up the song’s pickup pitch G properly on a right foot.⁵⁰ The return of the refrain brings about the same grouping change discussed above, but its repeat to end the song (when the original refrain lyric is returned to the melody) acts now as a corrective, the final melodic arrival on a *clos* ⁵¹ (G in m. 8) now coinciding with a firm stamp of the left foot.⁵¹ The text of V5 is redolent with the sweet

⁵⁰ I maintain imperfect major grouping throughout the melisma of free eighth notes in m. 11. The first beat passes through a third while the second decorates local leading tone C♯ with a lower neighbor.

⁵¹ The complete form with all its repeats is not represented but can be inferred from the example.

sorrow of separation. However far the noble lady may be, the refrain assures that she is always in the lover's thoughts.

As proposed above, the single pickup note of Virelai 5 (G) is perhaps short enough for dancing to commence on the trochaic downbeat of the first full measure without a right foot preparation at the start. Because a full beat of anacrusis (two minim rests and one minim pitch) is notated in all sources (Mach A, C, E, G, and Vg), the dance steps conclude neatly. This same anacrusis beat *would* be marked with a right foot when the last refrain returns and the final cadence would land on the left, according to convention (formal repeat not shown in Example 12). Yet some of Machaut's virelais have more problematic returns at that final formal juncture where the a section (refrain with new text) meets A (refrain with original text). All but one of the many repeats of Machaut's "Se d'amer me repentioie" (V20), for example, upend the entrained footing of the prior section.

The examples so far have been danced at the *tempus* (beat) level of mensuration, stepping on each semibreve; that is, two steps per measure of 6/8. In my analysis of V20, I extend the dance step methodology to the broader *modus* level of mensuration. The parsing of *modus*, a measure of breves per long, requires an act of analytical interpretation when longs are not present in a melody's notation.⁵² This study continues modern analytical work on medieval meter by considering mensural levels to be expressed not only by discrete notated rhythms but by free rhythm that add up to a given mensural unit, and marking those with dance steps. For example, a song may start with three minims that can be mentally grouped as

⁵² See Maw 2004, 60, in response to Hoppin 1960.

a semibreve's duration, the central mensural value marked by one step of the dancer's foot.⁵³

To dance at the *modus* level is to step not on each beat (semibreve) but on each measure (breve), thus the steps even more closely reflect metrical grouping as described in this study. In **Example 13**, broken bar lines show breves (with beats counting *tempus* divisions) while full bar lines show longs (with measures counting *modus* divisions). The same kinds of grouping complications found at the *tempus* level above can be found at the *modus* and *maximodus* level, too. As I pursue the grouping of ever larger durations, what were once practicable dance steps are extended into an analytical tool that can address the same issues across hierarchically larger timespans. Grouping disruption at these higher mensural levels requires interpretation akin to how the construal of hypermeter in later European music rests upon larger congruencies across the notation of measures. Interpretive preferences like locating cadential arrivals on strong beats—secondly, on any beat rather than off—and situating stressed syllables on metrically strong positions while relegating final mute vowels to weak one can clarify broader rhythmic grouping in medieval French dance songs.⁵⁴

Dancers of Machaut's V20 might begin conventionally with the left foot, as shown in **Example 13**. As the melody cascades down an octave from high G to low, an inflected F# overshoots the final G to land on an open A (^2) on the left foot in m. 11. Machaut sets his

⁵³ When grouping free rhythm into larger durations, the analyst is often guided by melodic sense units that express consonant intervals through neighbor or passing motions, by text alignment, by cadential pull and resolution, and, when applicable, by blocks of harmonic identity.

⁵⁴ Maw 2013 provides a thorough analysis of text setting in the Machaut sources, with reference to their mensural and metrical implications, and reconstructs some principles for fitting words to music in his songs.

poem in units of two or four bars, for the most part, neatly expressing imperfect *modus*. But the regular pace of declamation established in the refrain, two syllables per bar, is disrupted by the first ending's two bars per syllable. If this F# leading tone resolved to G in m. 10, rather than overshot to A a measure later, the refrain would take a balanced shape of 4+2+4 bars. As a result of the first ending's melodic elongation to five bars (4+2+5), the dancer takes every step on the opposite foot when repeating the melody according to the lyric's *formes fixes*. The refrain's second ending, however, is shorter than the first by a measure and, balance restored, the closed cadence on final G (m. 14) also falls on a strong left foot. If both refrain endings were of the same length either the final cadence would be on the "wrong" foot or the dance would have to begin with a stationary right foot or pickup measure.

The B section sets forth with a right foot (m. 15) opening again on high G, recalling the A section, but only winding down to an open cadence on D (^5) in m. 23. Just as in the refrain, the feet are again reversed when the B melody repeats according to the *forme fixe*. This is due to a musical setting of 4+5, resembling the A section's 4+2+5 but with the "extra" bar being at the front of the group of five. There is an incipient melodic return of the refrain's opening four bars in mm. 20-23, the extra bar being the anacrusic m. 19. The way its C# leading tone overshoots D recalls how F# overshot G a fifth lower at the end of the A section. It seems fitting that only in the final return of the refrain, where a (same music, new text) is succeeded by A (same music, same text), should V20 return the same music and text to the same footing. The threads of the song thus tied together, it finishes.⁵⁵

⁵⁵ My transcription of V20 conforms with all but one Machaut manuscript in which this song is represented (Mach B, C, E, G, and Vg). It differs only from Mach A, which appends a breve (half) rest after the *ouvert* cadence on A in

modus: L R L R L R L R L R! L

8 De tout sui si con-for-te-e | que ja mais n'iert ho-ste-le-e | tri-stes-se n'e-smay |

13 8 en mon cuer, ain-cois a-ray | lie et jo-li-e pen-se - e, |

24 8 tant com je vi-vray. |

Example 14. Machaut's V38/32, "De tout sui si confortee" (cantus line of the refrain only);

marked for dance at the *modus* level and for phrase organization

The refrain of "De tout sui si confortee" is essentially comprised of two phrases—labeled α and β in the Example to distinguish them from formal labels A and B—varied in their repetitions melodically, rhythmically, and metrically. The analytical model of dance reifies Machaut's elastic approach to melodic repetition here, as the same melodic skeleton is embodied across longer or shorter time spans. Phrase α circulates upper and lower neighbors around the pitch C, ultimately cadencing on that same open $\wedge 5$ (mm. 4-5). Phrase α is prepared by what I take to be a pickup (m. 1) before the first downbeat. Commencing the dance on m. 2 best responds to the song's musical structure for the following reasons. First, doing so ensures

the same effect, with the benefit of placing the final F on a downbeat (breve 27), by interpreting perfections at the *modus* level, just as Ludwig implies by his 3/2 time signature within the context of 2/4 and 4/4.

that the melodic, textual, and agogic accent on the cadential C in m. 4 is corroborated with a left foot. Second, a very close repeat of phrase α occurs without the pickup in mm. 14-16. Third, when the refrain repeats the pickup will be taken with the right foot, preserving m. 2 as a strong left foot downbeat. Fourth, the transposable motive  is a recurrent one throughout Machaut's songs.⁵⁷ In this song it is used as an anacrusic windup targeting the C of m. 2, a modally structural pitch (the tenor) made sweeter by the prior accented passing tone B.⁵⁸

Phrase β consists of an essentially stepwise ascent from the song's final, F, to C and back to down again. Schematically, the tonally open phrase α is followed naturally by the closure of phrase β . While that is ultimately what happens, the detail of V38/32's melodic shape is a bit more complex. A summary sketch is presented as **Example 15** below, with phrases marked and cadences labeled O for open, C for closed. The song's final is emphasized by a stem.

⁵⁷ From the *virelais* alone see examples of this motive in V5 (mm. 4-5), V6 (mm. 15-16), V10 (mm. 14-15), V11 (mm. 4-5, 10-11, and 14-15), V12 (mm. 8-11 and 18-21, second endings), V15 (m. 18), V17 (mm. 8-9), V18 (mm. 9-10), V23/21 (m. 9), V26/23 (mm. 12-14), V29/26 (mm. 4-5), V31/28 (mm. 14-16, 25-26, and 31-32), and V37/31 (mm. 10-11 and 19-20), not to mention the many variants of its pattern.

⁵⁸ While emphasis of the pitch C enjoys a modal rationale here, the modern analyst of medieval music must always be aware of creating the tonal emphases they wish to find through the creative acts of transcription, such as the metrical fitting of bar lines and deploying *musica ficta* for leading tones. Leach 2000, 45-46, reviews these issues in reference to fourteenth-century song. As explored in this study, dance provides another parameter of medieval performance that may regulate, and at the same time may tempt, the modern analyst into placing this or that note on or off the beat. *Caveat saltator*.



Example 15. tonal summary of V38/32 (cantus of the refrain only); phrases and cadences marked

In its first occurrence, phrase β 's F-C-F scale is followed by an *ouvert* termination on $\wedge^2$ (mm. 11-12), postponing tonal closure for a later phrase. The return of phrase α (mm. 14-16) restarts the same melodic trajectory and is again followed by phrase β , to which is now appended a different *ouvert* ending ($\wedge^7$ in mm. 22-23). For the refrain's final phrase, β is stated alone and only now rests conclusively on its final F (mm. 27-28).⁵⁹

Recall in **Example 14** that the initial measure was taken as a pickup, the dual benefit of this metrical interpretation being the situation of the first cadence on a left foot downbeat (m. 4) and its final mute vowel on the weak foot thereafter (m. 5). If this first phrase in V38/32 is right, the remainder of the song is more ambiguous. The other four cadences of the refrain are evenly split as to landing on the left or right foot, with the final cadence—the one that

⁵⁹ The tonal closure achieved only upon repeat of phrase β (mm. 24-8) could have been achieved the first time (m. 22) by simply landing on F ($\wedge^1$) instead of on E ($\wedge^7$). The necessity of the repeat, in that case, may be attributable to text painting. Throughout the poem Machaut's protagonist scorns despair and, in mm. 18-22, dwells willfully upon thinking only happy thoughts across an extraordinary melisma of four measures. If the rate of text declamation here paralleled the first phrase β instead (mm. 6-12), the appendage of a third phrase β would be unnecessary.

ultimately closes the song—ending against convention on the right foot (m. 27). Whichever way one regards the opening measure, the melodic repetitions that represent the heart of the song are metrically inconsistent and cannot feel the same way when danced. The easy bodily entrainment witnessed in my first example (recall **Example 5**) is frustrated here in a way, I stress again, that cannot be seen in manuscript or current modern editions, nor can it be felt in a concerted performance, but may have been quite salient to the medieval composer and singing dancer.

In the following section I will speculate further about how the framework of dance can be extrapolated as a modern analytical tool not only to label a score but also to *feel* larger grouping congruences and irregularities in Machaut's danced songs. Just as the left foot accrues a sense of *thesis* (and the right of *arsis*) in music designed for caroling, every other left foot can feel weightier and thus parse the musical structure akin to the more modern concept of hypermeter. For example, those left foot downbeats that mark phrase endings are more salient than those that occur merely out of alternation with the right. **Example 16** is a foray into representing these different kinds of danced downbeats, which in mensural terms reflects the difference between *modus* (breves per long, all bar lines in the example) and *maximodus* (longs per maxima, only the unbroken bar lines). Both levels are shown here, and I retain L and R insofar as they represent strong (L) and weak (R) at this abstract level of virtual dancing. This analytical presentation is analogous to the layered dots in metrical hierarchy analyses of Lerdahl and Jackendoff, or the *pulse frameworks* of Graeme M. Boone more recently, and one can easily imagine smaller L/R steps on each beat at the *tempus* level below what is shown

here.⁶⁰ I find in the hierarchical organization of medieval French mensuration a similar sense of weightiness accruing over increasingly long time spans as multiple layers of beats coincide as metrical groupings. Dance qua analysis goes further in characterizing the beats of Machaut's virelais because they are not all equally weighted in principle, given the association of the left foot with thesis argued here.⁶¹ This norm provides a backdrop against which to interpret, to measure, and, moreover, to embody passages of a danced song being out of step, where that step is specially calibrated to each song's tonal and formal articulations. It is especially at the deeper levels of Machaut's music where metrical irregularity is most prominent,⁶² and so makes the extension of a simple dance pattern into an analytical tool worth pursuing. In this regard, beginning to dance only in m. 2, if begun well enough, does not produce genre

⁶⁰ See their introduction to metrical hierarchy in Lerdahl and Jackendoff 1983, 18-30. Boone 2000, 4, presents a similar "grid of pulse classes and pulse levels."

⁶¹ Lerdahl and Jackendoff 1983, 19, do quickly associate undifferentiated beats with strong/weak poetic accent, in which beats 1 and 3 of 4/4 meter are strong while 2 and 4 are weak. In the transcription of Machaut's virelais here, this corresponds to two measures of imperfect *tempus* danced L R / L R. Dancing at the *modus* level reveals the first left step to be stronger than the second—parallel to beat 1's comparative strength over beat 3—because that step is also left at the *modus* level while the second *tempus* left falls under a weak right foot in *modus*.

⁶² On flexible metrical grouping as valued characteristic of deep mensural levels in Machaut, see Boone 2000, Maw 2004, and Lavacek 2022. The analytical pursuit of dancing *modus* and *maximodus* here resonates with Lerdahl and Jackendoff 1983, 22: "this large level of metrical analysis [at 4-bars] is open to interpretation, whereas smaller levels are not."

appropriate results throughout.⁶³ Two cadences (boxed in the example), including the final one, are overlooked by the feet at this summary analytical level.

maximodus: L R L R L R L R L R
modus: L R L R L R L R L R

8 De tout sui si con-for-te - e | que ja mais n'iert ho-ste - le - e | tri - stes - se n'e - smay |

13 R L R L R L R L R L R
 8 en mon cuer, ain - cois a - ray | lie et jo-li - e pen - se - e, |

24 R L R L R L R L R L
 8 tant com je vi - vray. |

Example 16. dancing V38/32 at the *modus* and *maximodus* level (cantus of the refrain only);

cadence points circled

In order to align melodic accent more closely with dance pulses, the following reading of the same virelai interprets mensural expansions (*perfections*) at the *maximodus* level such that all five cadences fall on a step, still more often on the right but at least not in between steps. My occasional interpretation of a triple grouping is marked for special interpretation against the normative backdrop of duple mensuration in Machaut's virelais. Mensural expansion here is simply a group of three thoughtfully inserted into the regular duple entrainment native to caroling. While such an effect can occur anywhere, it is most stylistic and convincing either just

⁶³ Corroborating the argument made about the *modus* level above, the first measure really ought to be a pickup at the weightier *maximodus* level also or else only two of five cadences fall on a beat at all and both of them are right feet (not marked in the example).

before a cadence (as in mm. 8-10, 13-15, and 24-26 of **Example 17**, effecting a written out ritard) or in capturing a pickup measure before a prominent downbeat (as in mm. 27-29).⁶⁴ In this way, **Example 17** offers a more flexible danced analysis for comparison with those regular steps that did not well match the phrase structure (viz. cadences) of **Example 16** above. Closely allied to those cadences, I note that the two oxytonic rhymes on *-ay* (mm. 11 and 27) are brought into alignment with the left foot in a strong metrical position. While the final pitch of the paroxytonic *pen-SE-e*, an ouvert E pointing towards the final F to come, also falls on a left foot (m. 22), the longest melisma in the song bestows ample stress upon its penultimate syllable (mm. 19-21).

⁶⁴ Other modern scholars have pointed to how the broadening of mensuration can achieve an effect like a ritard before cadence. Adjusting *modus* and larger mensural groupings to align harmonic units and cadences is central to the analysis of Machaut's motets in Lavacek 2022. Kohn 1981, 41–2, comments upon a cadential retardation effected through mixed meters in the Dunstable motet Sancta Maria. Desmond 2018, 217 (Example 6.5), shows a change in all voices from imperfect to perfect *modus*, which acts like a cadential retardation, at the conclusion of each *talea* of the Fauvel motet "Garrit/In nova." Maw 2004, esp. 64–5, locates *modus* perfections in an imperfect context leading up to cadences in Machaut's ballade "Gais et jolis" (B35), among other examples. He later (86) calls this effect "a reverse hemiola at the cadence." Further examples from Machaut's virelais are in Maw 2002, 84, Example 9b.

maximodus: L R L R—2—3 L

α β

8 De tout sui si con - for - te - e | que ja mais n'iert ho - ste - le - e tri - stes - se n'e - smay |

R—2—3 L—2—3 α β R—2—3 L

13 8 en mon cuer, ain - cois a - ray | lie et jo - li - e pen - se - e, |

R—2—3 β L—2—3 (repeat of A)

24 8 tant com je vi - vray. |

Example 17. dancing V38/32 flexibly at the *maximodus* level (cantus of the refrain only)

I have interpreted mensural expansions six times in this refrain. The first occasion is in mm. 8-10, where the right step is extended for three counts (notated R—2—3) before bringing down the left to mark the cadence on G in m. 11, an important melodic pitch that fell between steps in **Example 16**. Delaying the foot on a long drag triplet before the next downbeat represents a broadening out before cadence, the opposite effect of dancing a quick hemiola shown earlier in V14 (recall **Example 9**; three little steps in the place of two). The next danced perfection is interpreted across mm. 13-15 for the same end; to unite the conventional left foot thesis with a melodic arrival point, the cadence on C in m. 16, whereas this cadence came on the right foot when danced straight. Measures 19-21 and 24-26 similarly prepare their cadences, the first an open ending on E ($\wedge^7$) and the last closing the refrain on the song's final, F, both properly tied to the left foot. The perfection interpreted in mm. 16-18 is an expedient that sets up the usual cadential broadening in mm. 19-21, which preferably begins on the right foot so as to land a cadence on the left. The guiding intent for all these interpretive metrical

adjustments is for bodily entrainment to align with the music more conventionally, providing consistency in how the steps approach cadence and in the maintenance of left foot downbeats found wanting when this song is danced straight (**Example 16**).

The final measures of V38/32's refrain would not wrap around evenly on repeat—assuming the first measure is an undanced anacrusis—without implying another *modus* perfection (mm. 27-28-1). This final expansion is shown in parentheses running off the end into a repeat of the refrain that occurs at the end of the virelai form. In fairness, V38/32 does wrap around neatly if one simply walked through this song without regard for cadential placement. The methodology here, however, advocates for a parsing more sensitive to the tonal articulation of a song.

I have explored the cross-footed effects latent in this song as an act of modern critical reception, considering some of the practical as well as analytical consequences of dancing “fashionable *ars nova* rhythms” (Earp 1991, 125).⁶⁵ More than just walking in time, a dance that embodies grouping changes and embraces irregularity across the form allows participants to interpret the music and its text. To dance Machaut's V38/32 flexibly is to take action towards ensuring harmonious union between melodic and metric grouping parallel to the unswerving loyalty pledged by the courtly lover of its poetry.⁶⁶

⁶⁵ Earp 1991 traces the development of the danced song genre as a generic mixing of high courtly poetry with low metrical dance.

⁶⁶ In the first verse of V38/32, for example, the singer pledges that to do rightly and with a happy heart means everything, and that he shall do nothing less till death. [*Bien faire et avoir cuer gay, c'est tout; plus n'emporteray, quant seray finee;*]

The cantus may carry the melody here, but it is not the complete song. Restoring the tenor allows an evaluation of harmony and counterpoint in relation to metrical grouping and dance. The polyphonic analysis of **Example 18** pursues the flexible reading above as more responsive to V38/32's tonal organization.

maximodus: L R R L R R — 2 — 3 L L
modus: L R L R L R L R L R! L

8 De tout sui si con - for - te - e | que ja mais n'iert ho - ste - le - e tri - stes se n'e - smay |

8 TENOR

α β

5 3 5 6 8 6 8 5 3 1 3 3 5 8 3 3 1 3

R — 2 — 3 L — 2 — 3 R — 2 — 3 L L R
 R L R L R L R L R

13 en mon cuer, ain - cois a - ray | lie et jo - li e pen - se - e, |

α β

3 3 1 6 8 6 8 5 4 1 3 3 5 8 3 3 3 3

R — 2 — 3 L L
 L R L R L

24 tant com je vi - vray. |

β

5 8 5 3 1

Example 18. Machaut's V38/32 (refrain) with dance steps at two levels, phrase organization, and discant reduction

The note-against-note discant reduction at bottom reveals relationships between pitch and metrical grouping.⁶⁷ For example, Machaut is consistent in employing imperfect consonances on (*modus*) downbeats only in the approach of cadences, where their strong resolution tendency joins contrapuntal propulsion to the mensural broadening (*maximodus* perfection) I have suggested at mm. 8-10, 13-15, and 19-21.⁶⁸ Machaut's counterpoint against recurring phrases α and β —recalling that a free tenor voice is composed against the fundamental cantus in secular song, the reverse of sacred cantus firmus style—also shows a high degree of consistency. In both occurrences, phrase α begins with a stable fifth expanding into an unstable sixth, which drives out by stepwise contrary motion to the nearest stable interval, the perfect octave. This 6-8 directed progression occurs twice in descending sequence in each statement of phrase α , while dissonant passing sevenths provide even greater drive for resolution on C (mm. 4 and 16). The first two occurrences of phrase β are also very similar, cadencing each time on the same open dyad E/G that yearns for resolution on F (mm. 11 and

⁶⁷ My discant reduction was largely produced by the simple omission of dissonances against the tenor, especially when such notes occur off the beat in passing or neighboring melodic motion. Leach 2000 provides an instructive example of modern reductive analysis grounded in medieval theory.

⁶⁸ On the crucial concept of directed progressions in fourteenth-century polyphony, the classic source is Fuller 1992. Essentially, imperfect consonances are felt to drive towards the greater stability of perfect ones through stepwise contrary motion, especially $M3 \rightarrow P5$, $M6 \rightarrow P8$, and $m3 \rightarrow P1$. Polyphonic simultaneity is very tightly linked to consonance/dissonance placement in the mensural theories of John Garland, Franco of Cologne, and Anonymous IV, as detailed in Fuller 1981.

22). The final iteration of β (mm. 24-7) appears in compressed form and is harmonized differently. Replete with perfect consonance, a judicious third is employed only at the final cadence (3-1). Note the special contrapuntal strength of this final arrival, converging in contrary motion down a catalog of the harmonic consonances: 8-5-3-1 (mm. 25-28). This more solid setting of phrase β perhaps represents the singer's resolve to remain loyal "as long as I live" [*tant com je vivray*].

Among Machaut's *chansons baladées*, four are set in perfect *tempus*: V26/23, 27/24, 31/28, and 32/29. While dancing with two feet maps neatly onto binary groupings, repeatedly reinforcing one foot as strong and the other weak,⁶⁹ dancing three beats per measure (semibreves per breve) with the same two feet results in a feeling of rotation through metrical accents. That is, rather than every metrical accent pairing with the same foot, stepping on each semibreve causes an alternation of feet on the musically accentuated beat 1.⁷⁰ However, to the extent that there was a generic convention for left-footed duple accents, that norm may still be experienced as shifted to the next higher mensural level, that of *modus*. Because Machaut's triple *tempus* dances proceed LRL / RLR / LRL etc., the accrual of left foot downbeats (every two measures) creates an embodiment of duple *modus*, often ratified by tonal groupings. The crucial difference for dancing is that not every left foot is equally strong in perfect *tempus*, but every other left—marching along in imperfect longs—will feel like downbeats. To keep social

⁶⁹ Insofar as Mullally 1998, 19, insists that "social dance of the period requires music in duple time," these few songs in triple time may represent the artistic extension of a genre with roots in practical dancing.

⁷⁰ A potential alternative to this—marking triple groups as an unequal binary—maintains the usual L/R alternation per measure.

dancing most accessible, however, the few *chansons baladées* in triple time may instead encourage dancing at the *modus* level rather than *tempus*.

I will address two pertinent consequences of this. **Example 19** presents the opening of two of Machaut's virelais with dance notation at two levels, *tempus* below and *modus* above. While the alternation of which foot comes down on beat 1 may feel unnatural, given the preponderance of virelais in imperfect *tempus* (88%), the characteristic correlation of a dominant foot with tonal articulations is actually left undisturbed. All directed progressions in V27/24 and V32/29, except a single interior one, fall on the same foot. Moreover, this is true regardless of whether these songs are danced at the *tempus* or *modus* level because virelais in triple time, unlike their duple counterparts, align steps at both of these mensural levels. The difference is hierarchical, as the figure makes clear, with practical implications for tempo of the dance. Because the step on every beat 1 aligns, the decision to dance quickly (stepping on the semibreve beat, likely singing at a slower tempo) or more slowly (stepping on the breve measure, likely singing at a faster tempo) has no impact on the embodiment of tonal organization in virelais of perfect *tempus* so long as imperfect *modus* is left undisturbed. Take the first cadences of V27/24 and V32/29, for example. In V32/29, the directed progression to G in m. 4 falls on the right foot regardless of mensural level, as do all the others that follow, including the formal closure of each section. The case is nearly the same in V27/24, where only one internal, inverted directed progression (to F in m. 21) falls on the left foot while thirteen others fall on the right. The consistency of this result is highly suggestive of dances that should

begin with a pickup measure (breve), what Maw dubs anacrusic *modus*.⁷¹ The effect is to move all tonal emphases to the left foot, in line with generic convention established above. The inflected ascent of both melodies represents an introductory runup to the structural downbeat in m. 2 to be taken with a firm left foot. The lyric declamation of each of these openings, both seven-syllable lines, is typically trochaic. Yet Machaut’s rhythmic and melodic setting of each m. 1 overwrites this pattern with iambic in V32/29 and pyrrhic in V27/24, further conveying a sense that the structural downbeat of each song comes in the second measure. I suggest, then, that these dances may begin with a right foot, some prefatory dance gesture such as a plié, or with no step at all.

The image shows two staves of musical notation in 3/4 time. Above each staff are rhythmic markings: R?, L, R, L, R, L, R, L. The first staff is for the melody 'Da-me, mon cuer em - por - tes,' and the second is for 'En mon cuer a un des - cort'. The notation includes treble clefs, key signatures (one sharp for the first, one flat for the second), and various note values (quarter, eighth, and half notes) with rests. Vertical bar lines are placed at the end of each line.

Example 19. beginning of Machaut’s virelais “Dame, mon cuer emportes” (V32/29) and “En mon cuer” (V27/24); cantus lines only

⁷¹ Table 7 in Maw 2004, 94-5, provides a detailed list of *modus* organization in Machaut’s *formes fixes*, including flexible readings of perfect/imperfect alternation. Consult Table 1 of Maw 2002, 76, for a complete list of mensurations, including anacruses, and cadential plans in Machaut’s virelais.

An annotated score of the cantus line of V27/24, omitting tenor accompaniment, is given below as **Example 20**. All directed progression arrivals are circled. Because this song is also measured in perfect *tempus*, dance steps will be considered at the imperfect (binary) *modus* level, offering at least their analytical value in parsing medieval meter if not in directing medieval feet.

A *modus*:

En ma cuer a un des - cort | qui si fort le point et mort | que, sans men-

- tir, | s'A-mours par son doulz plai - sir | n'i met ac - cort | a-veuc ma da-

me, pour mort | me doy te - nir. | C'est de mon loy - al De - sir | qui me

wet fai - re ge - hir | le mal que port. |

Example 20. annotated score of Machaut’s V27/24, “En ma cuer;” cantus line only

By approaching the song this way, dancers can maintain the generic correlation of left foot strong beats. As mentioned above, the only directed progression that falls on the right foot is a weak, inverted one ($m3 \rightarrow 1$) in m. 21 (boxed in gray), which provides an anticipatory taste of the final F. The F of m. 21 is tonally, metrically, and lyrically subordinate to the stronger $M6 \rightarrow 8$ directed progression that closes the A section on the left foot in the next measure.

It is only conventional that each section of this song, A and B, should close with cadences on the left foot. Ending on the left prepares all of this *chanson baladées'* formal junctures to begin with a right, further reason to infer opening the dance on a right foot. Recall that when dancing at the *tempus* level, a left foot cadential downbeat is followed by a right foot offbeat before another left foot downbeat begins the next phrase.

The consistent metric placement of preparatory gestures can provide a musical cue for proper dance steps, not only at the beginning but as found throughout a danced song. In V27/24, the same gesture in m. 1, identified for its melodic directedness and unusual pyrrhic setting, returns often across the song, as at mm. 5, 11, 15, 17, 27, and in varied form at 15 and 23. That in every instance this melodic preparation falls on the right foot confirms the subordinate role of that foot and argues strongly for beginning this dance with a right or other preparatory gesture, as speculated above. In V31, the opening figure of m. 1 does not appear so melodically driven as the ascending leading tone of Virelais 27/24 and 32/29. Its function becomes clear through Machaut's consistent use of it as cadential preparation. The formal cadences that close both the A and B sections (mm. 21-22 and 34-35) are motivic repetitions of m. 1, as summarized in **Example 21**. Insofar as cadential arrivals are conventionally marked by a left foot, their preparatory gesture should be taken with the right, and this norm can be backdated to establish the beginning of a dance.

(R) L 21 R L 34 R L

8 Plus du - re a - mi - tie. sous-ri - ant.

Example 21. weak beat prep to strong cadential arrival in Machaut's V31/28, "Plus dure que un dyamant," mm. 1-2, 21-22 (end of A), and 34-5 (end of B); cantus only

The final danced song in triple time, "Tres bonne et belle" (V26/23), puts *modus* dancers off-kilter regardless of how it is begun. The cantus melody opens with the descending pentachord G-C, decorates that C (^1) with both neighbors (D in m. 2 and B in m. 3), then descends again towards G (mm. 4-5) to start the next phrase. A compressed variation of this melodic framework returns within the A section (mm. 16-19), reflecting the poetry's rhyme scheme (-*ueil*). But this melodic return comes against the prevailing metrical (*modus*) entrainment. Not only is the melody now metrically unstable, as felt through the dance, but it is also recontextualized by a different accompaniment in the two lower voices. **Example 22** presents these two passages in question for comparison, with steps at the *modus* level. The embodied metric disturbance here may shed light upon the courtly love lyric. While the lover's eyes are smitten with the beloved in the first phrase (*Tres bonne et belle, mi oueil joyeuse pasture prennent en vostre figure*), we know the heart of the second phrase will be stung by the pains of unrequited love rather than enjoy any sweet nurturing (*et mes cuers en vostre accueil vie et douce norreture*).

1 L R L R L

8 Tres bonne et bel - le, mi oueil Joy-

CONTRATENOR

TENOR

6 — 8
3 — 5

15 R L R L

8 Et mes cuers en vostre ac - cueil

6 — 8
3 — 5

Example 22. melody recontextualized in Machaut’s V26/23, “Tres bonne et belle;” mm. 1-5 and 15-19, directed progressions marked

The uneasy footing I am suggesting rests upon the ideal of undisturbed imperfect *modus* groupings. The directed harmony setting the preparatory word “et” that begins the second phrase (m. 15) suggests a right foot rather than a left. If left were to come down on m. 16, the perfect resolution of that imperfect harmony would kick off this phrase more parallel to the first. If this step alignment would be to advantage, the dancer may be tempted to interpret a *modus* perfection of measures 13-15, a triple group whose final measure is the preparatory 15

and which all the more forcefully begins the second phrase at m. 16 on a proper downbeat in all parameters. However, mm. 13-14 form a clear cadence (E6/3 to D8/5) which ought to proceed right—left by the same logic. This in turn pushes back an initiating left foot to m. 12 and the *modus* perfection somewhere between mm. 9-12, an exercise left to the reader. In any event, the entire refrain cannot repeat on the same dance footing (at the *modus* level) when it returns at the end of the virelai form because it is an odd number of measures. If the final cadence (m. 23) is to be a left foot, respecting this as the very strongest convention, m. 1 really ought to be a right and something in the middle will have to be perfected.

This study does not presume to answer how Machaut's songs were danced, only to grapple with some questions that arise when pairing regular dance steps with irregular musical groupings. A polyphonic and metrically complex song like Machaut's Virelai 38/32 stands out as an art song evolved from a dance tradition it no longer exemplifies.

I have applied the framework of simple round dancing to interpret metrical organization in some secular medieval songs known as a genre, and in some cases individually, to have been danced. While simple left-right alternation fits many works designed for dancing by their very nature, the freer blending of duple and triple groupings at various mensural levels found in some *chansons baladées* of Machaut—a composer also of complex motets, a genre Johannes de Grocheio famously claimed for the learned—perhaps elevated dance style off the dance

floor.⁷² The analytical framework developed here was prompted by the *carole*: begin with the left foot, alternate feet, and try to cadence on the left. Its simple expectations thrust to the fore the kinds of mensural and metrical sophistication that contributed to a shift from physical to virtual embodiment in how Guillaume de Machaut's danced songs were encountered in the fourteenth century and can guide fresh interpretation—and, indeed, dancing—in our own.

MACHAUT SOURCES

A: Paris, Bibliothèque Nationale de France, fonds français 1584

B: Paris, Bibliothèque Nationale de France, fonds français 1585

C: Paris, Bibliothèque Nationale de France, fonds français 1586

E: Paris, Bibliothèque Nationale de France, fonds français 9221

F-G: Paris, Bibliothèque Nationale de France, fonds français 22545-6

Vg: Ferrell-Vogüé, private ownership of James E. and Elizabeth J. Ferrell

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⁷² Page 2000 discusses some of the fourteenth-century motet's elevated characteristics in relation to Grocheio's witness, including a bolder usage of dissonance, polytextual assonance, learned counterpoint, and some suggestions for performance practice. More comprehensive discussion may be found in Page 1997, chapter 3.

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